Distill: Keep the Knowledge, Lose the Weight

Ericsson Build Hackathon 2026 — Data Storage Optimization

Team Distill

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Abstract

Distill reduces the storage used by Ericsson's recordings and documents without losing the information inside them. Videos are turned into slide pictures, a speaker-attributed transcript and short voice samples; official documents are kept as originals, copies become references and older revisions are stored as their changes only. On Ericsson's 12 sample files, total storage fell from **1.45 GB to 25.0 MB (98.3%)**. Everything runs offline on a laptop CPU.

1 Architecture

Distill scans three sources, fingerprints every file, and processes videos and documents in two pipelines. Results are stored in an SQLite database and a content-addressed object store (each picture stored once). A FastAPI backend serves them to a Next.js user interface.

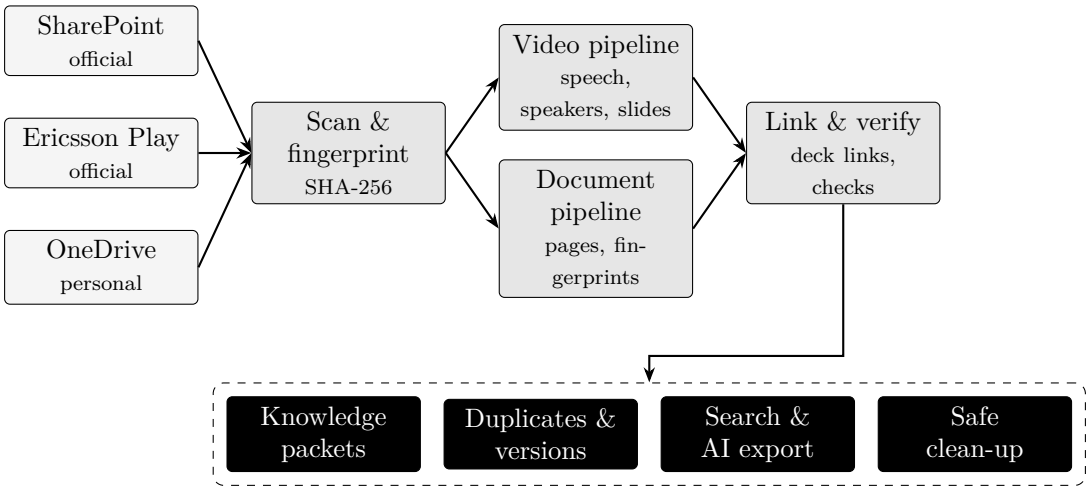


Figure 1: Distill architecture. Official sources are SharePoint and Ericsson Play; OneDrive is a personal drive.

2 Tools and models

Only four AI models are used; everything else is a deterministic algorithm. All run offline.

Tool	Type	Used for
Whisper base (faster-whisper, int8)	AI model	Speech to text with word timestamps
pyannote 3.0 + WeSpeaker (sherpa-onnx)	AI model	Who spoke when (speaker detection)
RapidOCR (PaddleOCR, ONNX)	AI model	Reading text on slide pictures and PNG images
all-MiniLM-L6-v2	AI model	Search by meaning
FFmpeg, OpenCV	Algorithm	Reading video, audio extraction, slide-change detection
Opus (16 kbps), WebP (q70), gzip	Algorithm	Compressing audio, pictures and text
SHA-256	Algorithm	Exact fingerprints of files and slide texts
pHash / dHash	Algorithm	Picture fingerprints (look-alike slides, duplicate videos)
PyMuPDF, python-pptx, LibreOffice	Library	Reading and rendering PDF and PowerPoint pages
olefile, XML parsing	Library	Reading e-mails (MSG) and Word files (DOCX)
SQLite + FTS5	Database	Storage and keyword search
FastAPI, Next.js	Web	Backend API and user interface

Table 1: Tools and models.

3 How each file type is compressed

3.1 Videos

1. **Fingerprint** the original (SHA-256) so it can later be proven unchanged.
2. **Extract audio** (FFmpeg) and **transcribe** it (Whisper): every sentence with start and end time.
3. **Detect speakers** (pyannote + WeSpeaker) and keep an 8-second voice sample per speaker.
4. **Detect slide changes** (OpenCV), ignoring the webcam area; keep one picture per slide, taken just before it changes.
5. **Read each slide** (RapidOCR) and **link** it to the official deck page (75% word overlap + 25% picture similarity, threshold 55%).
6. **Package** the result: Opus audio, WebP slide pictures, gzipped transcript and notes.

The package can be stored at five levels. Each level is the sum of the files it keeps:

Level	What is kept	Meeting	GraalVM	Jury
1 Original	Full video	1166.39	210.27	84.55
2 Smaller video	Re-encoded: H.264, 5 fps, max 1280 px	157.75*	5.70	13.84
3 Audio + slides + text	Opus audio + slide pictures + transcript + voices	24.07	2.21	1.44
4 Slides + text	**Level 3 without the audio (default)**	**7.47**	**0.62**	**0.94**
5 Text + links	Level 4; linked slides replaced by links to the deck	5.49	0.62	0.94

Table 2: Size per level in MB (1 MB = 1024 KB). *Estimated: Level 2 is not created for videos longer than 25 minutes.

3.2 PDF and PowerPoint

Every page is read directly from the file (PyMuPDF, python-pptx) and drawn as a picture (LibreOffice for PowerPoint). Each page gets a text fingerprint (SHA-256 of the cleaned text, footers

and page numbers removed) and a picture fingerprint (pHash). Revisions are read from Ericsson's document control, e.g. `Uen | A | 2019-10-11`. Storage policy:

- **Official copy:** the original file is kept unchanged.
- **Exact copy elsewhere:** replaced by a reference to the official copy (0 B).
- **Older revision:** only the slides that differ from the newest revision are stored, plus references to the shared slides.

3.3 Word, e-mail and image files

Word (DOCX) and e-mail (MSG) text is extracted for search; the official Word file is kept, the e-mail is stored as compressed text. Images (PNG) are fingerprinted with SHA-256 and their text is read with OCR; an exact copy becomes a reference.

4 Results per file

#	File	Source	Initial MB	Final MB	Saved	Stored as
1	Brief.png	SharePoint	0.22	0.22	0.0%	Original
2	C2L Presentation (Rev B).pdf	SharePoint	3.38	3.38	0.0%	Original
3	Challenge_Ericsson.docx	SharePoint	0.27	0.27	0.0%	Original
4	Ericsson Response Intro.pdf	SharePoint	8.37	8.37	0.0%	Original
5	SD-WAN final.pptx (Rev B)	SharePoint	3.51	3.51	0.0%	Original
6	Meeting recording (155 min)	SharePoint	1166.39	7.47	99.4%	Slides + text
7	GraalVM talk (16 min)	Ericsson Play	210.27	0.62	99.7%	Slides + text
8	Jury and prizes (6 min)	Ericsson Play	84.55	0.94	98.9%	Slides + text
9	C2L Presentation (Rev B) (1).pdf	OneDrive	3.38	0.00	100.0%	Reference
10	Copy of Brief.png	OneDrive	0.22	0.00	100.0%	Reference
11	SD-WAN final2.pptx (Rev A)	OneDrive	3.30	0.20	94.0%	Delta (7 + 18 ref.)
12	Chat history.msg	OneDrive	0.08	0.00	99.2%	Text
	Total		**1483.94**	**24.99**	**98.3%**	

Table 3: Initial and final size per file (1 MB = 1024 KB). A final size of 0.00 is a reference of a few bytes; file 12 is 656 bytes. Delta = 7 changed slides stored + 18 references to Rev B.

Quality checks on the meeting recording: 95.8% of on-screen slide text is found in the kept slides, 90% of checked moments match by direct picture comparison, and OCR reads 95.1% of the words of 36 official slides correctly. The original files are never changed (verified with SHA-256).